POWER AUTOMATION TEAM

DCS & INSTRUMENTS SPECIALISTS



COMPANY PROFILE 2026

THE TEAM THAT BRINGS YOUR PLANT CONTROL SYSTEM ONLINE.

DCS installation supervision · loop checks · commissioning · start-up · training

EGYPT · SAUDI ARABIA · OMAN — SINCE 1991

PAT AT A GLANCE

THE COMMISSIONING SPECIALISTS

Since 1991, EPC contractors and control-system vendors have called PAT when a distributed control system has to be brought online: installation supervision, IO loop checks, device tests, commissioning, start-up, punch-list closure and operator training — in power plants across Egypt, Saudi Arabia and Oman.

1991	28	3	24/7
Established	Reference projects	Countries	Support



PAT commissioning team on site.

WHO CALLS US

EPC contractors and system vendors — Emerson, Hanwha, Daewoo, Skoda, CMI, INITEC, GE, Orascom, Saudi FAL, KBR.

WHEN

New DCS projects, migrations and retrofits, outages, start-ups, loop tests and punch-list closure.

WHAT THEY GET

Engineers who have done it before, procedures that are safe and easy to track, and a team that stays to commercial operation.

ABOUT US

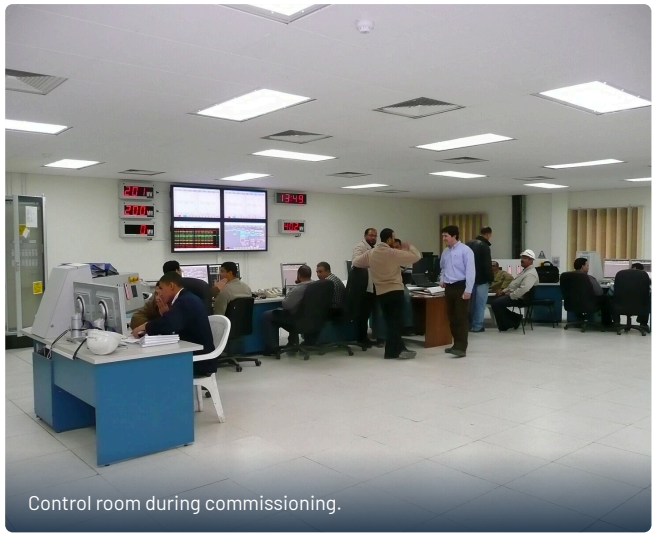
THIRTY-FIVE YEARS ON SITE

Power Automation Team (PAT) was established in 1991 to cover power plants in Egypt and the Middle East. Our work spans power generation, water and petrochemical plants.

The team consists of qualified local engineers and technicians who have worked with major international companies, including Emerson, Techint, Alstom, GE, Siemens, Skoda, CMI, Iberdrola and INITEC.

We provide engineers and technicians on a contractual basis — part-time or full-time, for short or extended periods — screened to match each job, and we prepare and implement commissioning procedures adapted to each plant so that our clients deliver faster, safer and at optimum cost.

Ahdy Abu-Malwa
Founder & Managing Director — recognised by Emerson Process Management as DCS Lead Engineer on Emerson Ovation projects in Egypt.



Control room during commissioning.

VISION

To be the premier automation control-systems provider for power, petrochemical and industrial customers — quality, reliability and customer satisfaction first.

VALUES

Quality above all · Safety in every step · Transparency in communication · Teamwork · Continuous development

SERVICES

WHAT PAT DOES ON SITE

From the first cabinet to commercial operation.

DCS Engineering & Commissioning

Installation supervision, software and hardware configuration and commissioning of distributed control systems — mainly Emerson Ovation — in power and industrial plants.

- Removal of old panels and supervision of the new DCS installation
- Software and hardware modifications and system configuration
- Database, logic and graphics engineering ready for FAT

PLC Programming & Integration

Engineering, installation and commissioning of PLC-based systems, including Siemens S7, and troubleshooting of PLC controls.

- PLC engineering, software and hardware installation and commissioning
- Siemens S7 IO loop and device checks
- Troubleshooting of PLCs, switchgear and emergency systems

Commissioning & Start-up

Commissioning teams for gas turbines, steam turbines, boilers, HRSGs and balance of plant, helping EPC contractors deliver the plant faster, safer and at optimum cost.

- GT, ST, boiler, HRSG and BOP commissioning and start-up staff
- Commissioning procedures prepared and implemented for each plant
- Reliability tests and return of units to commercial operation

I&C and Electrical Commissioning

Loop checks, device tests and calibration for instrumentation and control systems, plus electrical commissioning, to ensure safe and efficient operation.

- IO cold and hot loop tests
- Plant device functional tests
- Field instrument calibration

Training & Development

Customized operation and maintenance training for owner staff, including DCS operation and simulator instruction.

- Owner staff courses on DCS plant operation and maintenance
- Training instructors for power plants and DCS simulators
- Training academy set-up and course material for utility engineers and technicians

Technical Staffing & Manpower

Screened engineers and technicians placed with contractors and owners on a contractual basis — part-time or full-time, for short or extended periods.

- DCS Ovation hardware and software engineers
- Field instrument engineers, fitters and technicians
- Mechanical, electrical and I&C commissioning engineers

Troubleshooting & Emergency Support

Rapid support for routine and critical problems that affect your facility, available 24/7.

- DCS Ovation and DeltaV troubleshooting alongside the vendor team
- Gas turbine start-up and operation troubleshooting
- Generator, switchgear and PLC troubleshooting

One team, one schedule

Any combination of these services, staffed and supervised by PAT for the duration of your outage or start-up.

+20 109 991 2020
info@patdcs.com

HOW WE WORK

WHERE PAT FITS IN YOUR PROJECT

Call us for one step or for the whole sequence — the same engineers stay from installation to hand-over.

01

Installation supervision

Removal of old panels, new cabinets, cabling and power-up of the DCS.

02

Configuration

Database, logic and graphics checks; hardware and software modifications.

03

Loop checks & device tests

Cold and hot IO loop tests, device functional tests, instrument calibration.

04

Commissioning

System-by-system commissioning of GT, ST, boiler, HRSG and BOP with the plant team.

05

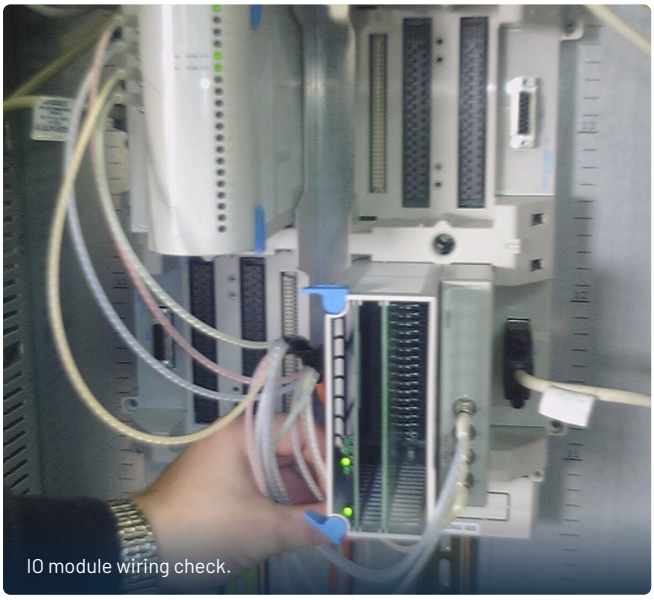
Start-up & reliability run

Start-up, troubleshooting and normalising the units to commercial operation.

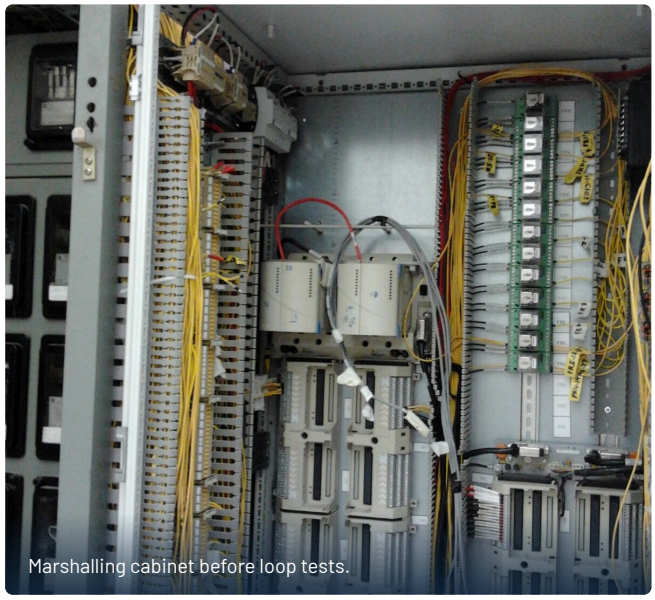
06

Hand-over & training

Punch-list closure, documentation, owner O&M courses and warranty support.



IO module wiring check.



Marshalling cabinet before loop tests.

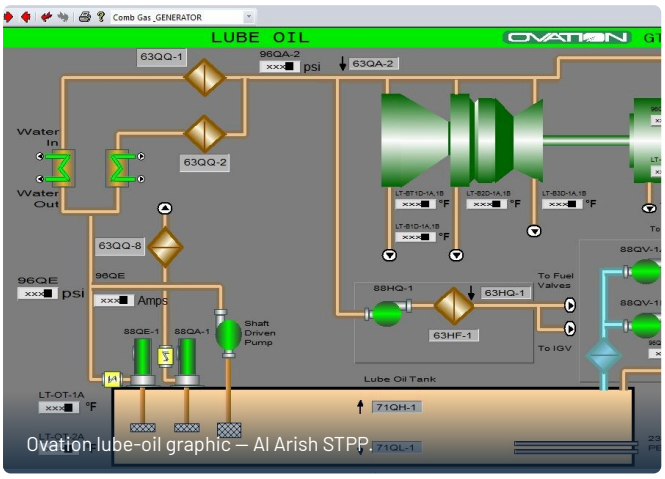
Our procedures are written to be safe, easy to implement and documented so that every loop and device is traceable.

CONTROL SYSTEMS

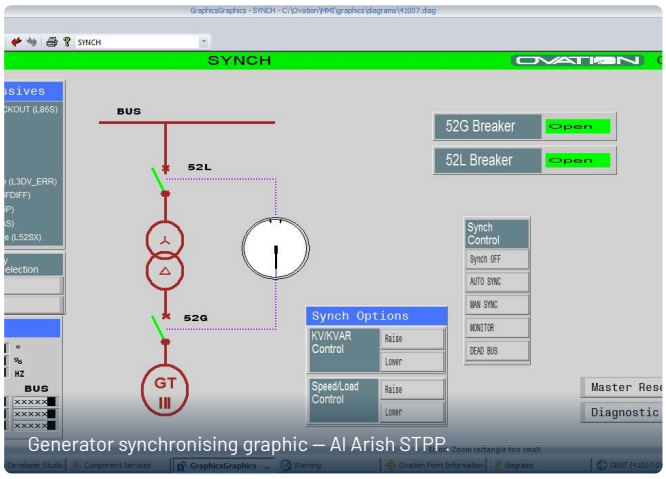
PLATFORMS WE COMMISSION

Most of our work is on Emerson Ovation; our teams have also commissioned and troubleshot the systems below.

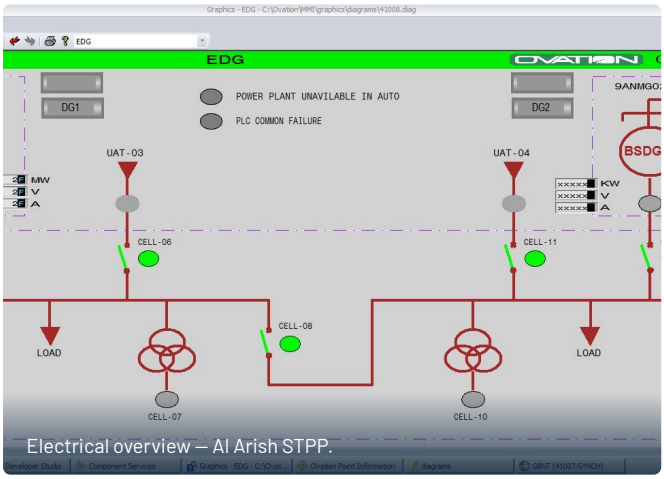
Emerson Ovation · Westinghouse WDPF → Ovation · ABB → Ovation · Emerson DeltaV · Alstom DCS · Siemens S7 PLC · GE gas-turbine controls · TCS → Ovation



Ovation lube-oil graphic — Al Arish STPP.



Generator synchronising graphic — Al Arish STPP.



Electrical overview — Al Arish STPP.



Ovation operator panel on a gas-turbine control cabinet.

CREDENTIALS

RECOGNISED BY EMERSON PROCESS MANAGEMENT



> *"In recognition of Ahdy Abu-Malwa for technical contributions and excellence in delivery of Emerson Ovation Expert Control System Solutions within the Arab Republic of Egypt."*

DCS LEAD ENGINEER

- New Talkha 750 MW CC Expansion
- Damietta 1,230 MW WDPF to Ovation Migration
- El-Kureimat II 750 MW CC Power Station

DCS ENGINEER

- Ataka 900 MW Steam Modernization

Signed by Robert Yeager, President, Emerson PWS, and T.C. Chan, Project Manager, Emerson PWS.

Emerson, Ovation, DeltaV and PlantWeb are trademarks of Emerson Electric Co. PAT is an independent company; these names describe work performed and recognition received.

SELECTED PROJECTS

REFERENCES THAT MATCH YOUR NEXT JOB

1,230 MW

DCS MIGRATION

Damietta CCPP 1,230 MW

Egypt · 2006 · Emerson Ovation · with Emerson

- Installation and commissioning of the new DCS parts
- Commissioning and start-up, returning the DCS units to commercial operation
- Owner staff training courses on DCS plant operation and maintenance

2,000 MW

OVATION DCS — APS CHECK & COMMISSIONING

Sur IPP-CCPP 2,000 MW

Oman · 2014 · Emerson Ovation · with Emerson, Daewoo

- Check, troubleshooting and modification of the APS (Automatic Plant Start) software so it works as one power block plant start
- Plant systems commissioning and start-up

1,300 MW

DCS MIGRATION

El Kurimat I 2×650 MW

Egypt · 2015–2016 · Emerson DCS · with Emerson

- Site management and supervision, inspecting and checking out the removal of the old DCS parts and the installation of the new DCS, on a very tight schedule
- Start-up and commissioning of the new DCS, powering up and configuring it for normal operation
- Plant device checks and tests (database, graphics, logic) and hardware and software modifications
- Cold and hot tests of plant devices and return of the DCS to normal commercial operation

SELECTED PROJECTS

CONTINUED

Ovation
FROM ABB

DCS REHABILITATION — ABB TO EMERSON OVATION

Shoaiba Power Plant, Stage I Phase 1

Saudi Arabia · 2021–2022 · Emerson Ovation · with Emerson, Giza Systems

- IO loop checks, device tests and commissioning of the boiler, turbine, BOP, desalination and demi water systems
- Troubleshooting of the Ovation DCS together with the Emerson team
- Comprehensive support throughout the project on a tight construction schedule to return the plant to commercial operation

12
PIBS

DCS UPGRADE — DELTAV

Abqaiq plants — DCS upgrade, PIB-2 to PIB-18 (Aramco)

Saudi Arabia · 2022–2023 · Emerson DeltaV · with Emerson, Saudi FAL, KBR

- Engineering document review and verification of cabinet and marshalling staging before FAT
- Supervision of construction activities
- Hot and cold test commissioning for each PIB shutdown
- Troubleshooting of the DeltaV system with the Emerson team

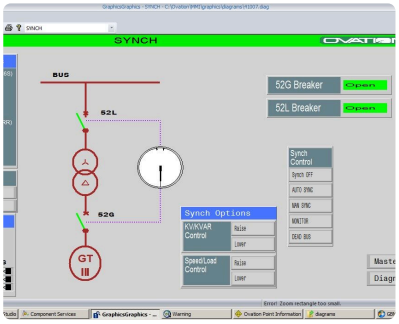
60
MW

DCS MIGRATION

Al Arish STPP 2×30 MW

Egypt · 2011–2012 · Emerson DCS · with Emerson

- Complete DCS removal and installation scope and migration activities
- Plant device check and test, commissioning of plant systems and return to normal operation
- Owner staff training courses on DCS plant operation and maintenance



PROJECT REGISTER

ALL 28 REFERENCE PROJECTS

Listed as in our reference list; contracting party in the last column.

NO.	PROJECT	SCOPE	COUNTRY	YEAR	SYSTEM	WITH
01	**Ataka Power Plant 2×300 + 2×150 MW**	DCS retrofit	Egypt	2004	Emerson Ovation	Emerson
02	**Damietta CCPP 1,230 MW**	DCS migration	Egypt	2006	Emerson Ovation	Emerson
03	**New Talkha & El Kurimat II CCPP 750 MW (two projects)**	DCS installation & commissioning	Egypt	2007–2009	Emerson Ovation	Emerson
04	**New Talkha & El Kurimat II CCPP 750 MW — BOP**	BOP commissioning & start-up	Egypt	2007–2009	—	Skoda
05	**New Talkha & El Kurimat II CCPP 750 MW — HRSG**	HRSG commissioning & start-up	Egypt	2007–2009	—	CMI
06	**Suez Sokhna Port United Sugar Project**	PLC commissioning	Egypt	2009	Siemens S7 PLC	Savola
07	**New Talkha CCPP 750 MW — CWP**	CWP commissioning & start-up	Egypt	2009	—	INITEC
08	**Shoaiba ST Power Plant 3×250 MW**	DCS retrofit	Saudi Arabia	2009–2010	Emerson DCS	Emerson
09	**PP4 Riyadh Power Plant — GE gas turbines 2×25 MW**	DCS retrofit	Saudi Arabia	2009	Emerson Ovation	Emerson
10	**El Kurimat III 750 MW**	I&C commissioning	Egypt	2010	Alstom DCS	INITEC
11	**Shoaiba lime water plant**	DCS engineering for FAT	Saudi Arabia	2010	Emerson DCS	Emerson
12	**Rabigh Power Plant 4×300 MW**	DCS upgrade	Saudi Arabia	2009–2011	Emerson DCS	Emerson
13	**Al Arish STPP 2×30 MW**	DCS migration	Egypt	2011–2012	Emerson DCS	Emerson
14	**Old Talkha 210 MW — TCS Unit 2**	DCS retrofit	Egypt	2011	Emerson Ovation	Emerson

PROJECT REGISTER

CONTINUED

NO.	PROJECT	SCOPE	COUNTRY	YEAR	SYSTEM	WITH
15	**Solar Energy Kurimat Power Plant**	Plant commissioning & start-up	Egypt	2011	—	Iberdrola
16	**SEC4A Al Jouf — Fiat gas turbines 7×35 MW**	DCS upgrade	Saudi Arabia	2009–2012	Emerson DCS	Emerson
17	**SEC4A Al Quryat — BBC gas turbines 2×22 MW**	DCS upgrade	Saudi Arabia	2013	Emerson DCS	Emerson
18	**Sur IPP-CCPP 2,000 MW**	Ovation DCS — APS check & commissioning	Oman	2014	Emerson Ovation	Emerson, Daewoo
19	**Rabigh II 2×300 MW**	DCS modernization	Saudi Arabia	2014	Emerson DCS	Emerson
20	**Marafiq Yanbu II 3×270 MW**	New DCS	Saudi Arabia	2014–2017	Emerson DCS	Emerson, Hanwha
21	**El Kurimat I 2×650 MW**	DCS migration	Egypt	2015–2016	Emerson DCS	Emerson
22	**Makkah Power Plant — GE gas turbines 6×65 MW**	DCS upgrade — GT3 troubleshooting	Saudi Arabia	2016	Emerson DCS	Emerson
23	**Al Qaseem Power Station — Westinghouse gas turbines 9×50 MW**	GT9 valve replacement & commissioning	Saudi Arabia	2017	—	Saudi FAL
24	**West Damietta Power Station CCPP — GT 4×125 + 1×250 MW**	I&C commissioning	Egypt	2017	—	GE, Orascom
25	**Yanbu I & II — Marafiq training academy**	Consulting & training	Saudi Arabia	2019	—	Leoron
26	**El Esraa petroleum tanker ship, Alexandria**	Electrical & PLC troubleshooting	Egypt	2020	—	—
27	**Shoaiba Power Plant, Stage I Phase 1**	DCS rehabilitation — ABB to Emerson Ovation	Saudi Arabia	2021–2022	Emerson Ovation	Emerson, Giza Systems
28	**Abqaiq plants — DCS upgrade, PIB-2 to PIB-18 (Aramco)**	DCS upgrade — DeltaV	Saudi Arabia	2022–2023	Emerson DeltaV	Emerson, Saudi FAL, KBR

WHERE WE WORK

THREE COUNTRIES, ONE TEAM

14
Egypt

Ataka Power Plant 2×300 + 2×150 MW	2004
Damietta CCPP 1,230 MW	2006
New Talkha & El Kurimat II CCPP 750 MW (two projects)	2007–2009
New Talkha & El Kurimat II CCPP 750 MW	2007–2009
New Talkha & El Kurimat II CCPP 750 MW	2007–2009
Suez Sokhna Port United Sugar Project	2009
New Talkha CCPP 750 MW	2009
El Kurimat III 750 MW	2010
Al Arish STPP 2×30 MW	2011–2012
Old Talkha 210 MW	2011
Solar Energy Kurimat Power Plant	2011
El Kurimat I 2×650 MW	2015–2016
West Damietta Power Station CCPP	2017
El Esraa petroleum tanker ship, Alexandria	2020

13
Saudi Arabia

Shoaiba ST Power Plant 3×250 MW	2009–2010
PP4 Riyadh Power Plant	2009
Rabigh Power Plant 4×300 MW	2009–2011
SEC4A Al Jouf	2009–2012
Shoaiba lime water plant	2010
SEC4A Al Quryat	2013
Rabigh II 2×300 MW	2014
Marafiq Yanbu II 3×270 MW	2014–2017
Makkah Power Plant	2016
Al Qaseem Power Station	2017
Yanbu I & II	2019
Shoaiba Power Plant, Stage I Phase 1	2021–2022
Abqaiq plants	2022–2023

1
Oman

Sur IPP-CCPP 2,000 MW	2014

WORKED WITH — CONTRACTORS & VENDORS

Emerson · Hanwha · Daewoo · Skoda · CMI · INITEC · Iberdrola · GE · Orascom · Saudi FAL · Giza Systems · KBR · Savola · Leoron

PLANT OWNERS

Saudi Electricity Company · Saudi Aramco · Marafiq · power plants of Egypt's national grid

TRAINING

COURSES FOR OWNER OPERATORS AND MAINTENANCE STAFF

Delivered at the plant, on the plant's own system, at the end of commissioning — and as stand-alone courses.

Ovation operator course — 4 days

Day 1	Typical Ovation DCS structure (theory)
Day 2	Alarm system and process diagrams
Day 3	Trend display and point information
Day 4	Point review, point viewer and error log



Ovation engineering workstation.

Ovation maintenance course — 10 days

Day 1	Basic operator functions; right-click functions in Ovation Developer Studio (OV 100)
Day 2	Creating and modifying analog, digital and packed points (OV 100)
Day 3	Monitoring, tuning and calibrating existing control (OV 200)
Day 4	Modifying existing control schemes (OV 200)
Day 5–6	Creating new control sheets (OV 200)
Day 7	Graphics Builder and non-point items
Day 8	Building point-related graphics
Day 9	Building and implementing control-interface graphics
Day 10	Building and using macros on main graphics

Also: GT, ST, boiler, HRSG and BOP operation & maintenance courses, DCS simulator instruction, and training-academy set-up (Marafiq, Yanbu).

ON SITE

OUR PEOPLE, IN THE FIELD

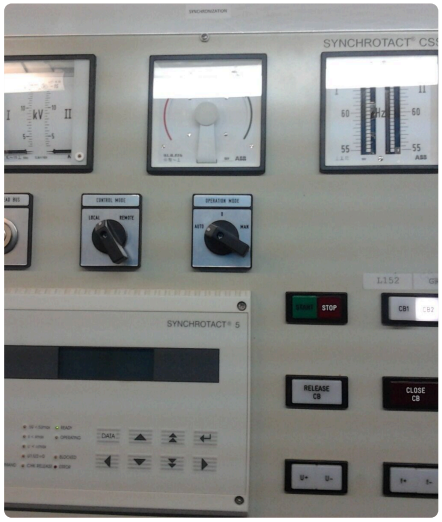




Night shift during start-up.









Start-up in the control room.

WORKING WITH PAT

HOW TO ENGAGE US

Scope package

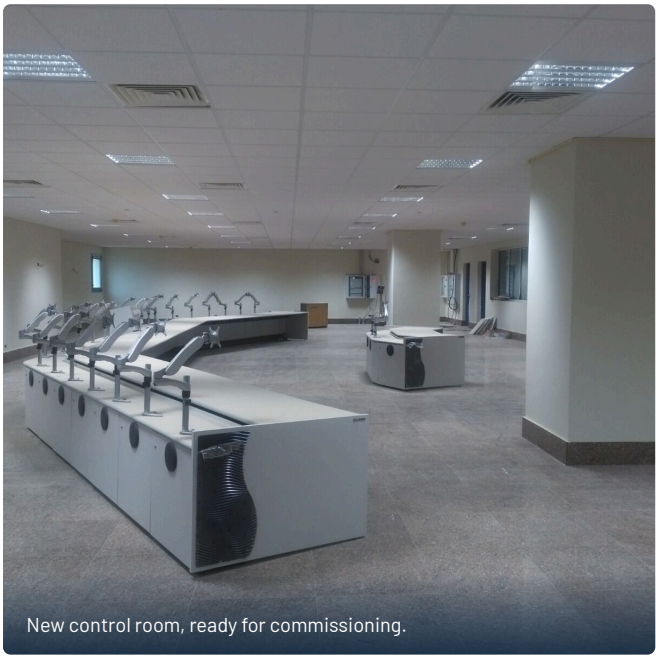
A defined scope — e.g. loop checks of a unit, DCS migration support, start-up of a GT — with our own supervision and procedures.

Technical staffing

Screened engineers and technicians in your team: DCS HW/SW, I&C, mechanical, electrical, operators. Part-time or full-time, short or extended.

Emergency support

A team on site fast for problems that affect plant availability — DCS, gas turbines, PLCs, switchgear. Available 24/7.



New control room, ready for commissioning.

Why contractors work with PAT

High-quality personnel

Engineers and technicians with long power-plant experience, screened for each job.

Competitive billing rates

Competitive pricing without compromising on quality.

Flexible manpower

The team grows and shrinks with your scope and schedule.

Commitment to quality

On time, within budget, documented and traceable.

Talk to us about your next outage or start-up

PHONE

+20 109 991 2020 (24/7)
+20 122 359 0402 (technical support)

E-MAIL

info@patdcs.com
sales@patdcs.com
service@patdcs.com

OFFICE

New Damietta, Second District,
adjacent to 6 Al Bahr Street, Egypt

POWER AUTOMATION TEAM

DCS & INSTRUMENTS SPECIALISTS

+20 109 991 2020 · info@patdcs.com

patdcs.com

باور أوتوميشن تيم